# Data Mining Analysis of Diabetes Disease Prediction

Course: Data Warehousing & Data Mining

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# CHAPTER ONE

## 1.0 Introduction

Looking at patient health data, this work explores how computers learn to predict diabetes. With millions affected globally, and projections pointing to rising fatalities, finding faster ways to spot the condition feels urgent. Early on, it became clear that machines might help doctors make sense of complex patterns hidden in routine tests. Instead of guessing, algorithms sort through numbers to find signals people miss. This opening section sets up what comes next - why prediction matters, what gaps exist in current methods, the core challenge we face, plus goals tied to training models that respond accurately when tested. Much depends on clean inputs; messy records confuse even smart systems. One step forward: teaching software to weigh factors like blood sugar or age without human bias shaping results.

## 1.1 Study Background

Nowadays machines help doctors care for people with long term health problems in smarter ways. Though diabetes shows subtle hints early on, old fashioned checks miss these clues too easily. Each day clinics collect piles of details like sugar readings, weight stats, blood force numbers, medicine doses, and patient years - all adding up faster than humans alone can sort through them.

Earlier, doctors used basic cutoffs to spot diabetes - like blood sugar above 126 after not eating - but those rules missed complex links among risks. Because body weight and aging interact in messy ways, so do genes and how cells respond to insulin; old methods overlooked that. Now, smarter math tools help uncover patterns hidden before, especially ones built on group strategies such as Random Forest or tree-shaped logic paths. Instead of relying on rigid lines, they adapt when faced with tangled health clues. A well-known collection of medical details from 1268 individuals - the upgraded Pima Indians Diabetes Database - is at the core here. Each person contributes eight real-world measurements tied to diagnosis. Different number-crunching styles take shape across this ground, tested side by side without favor. The aim? To see which path traces reality most closely.

## 1.2 Problem Statement

Even when spotting diabetes early matters a lot - stopping problems like heart issues, failing kidneys, damaged nerves, or losing sight - many clinics still struggle to diagnose it right every time. One big snag? Doctors often see blood sugar numbers or body mass index results in their own way. That means two people with the same test might get different labels. Some walk away misinformed or wait too long for answers, risking harm - all because the system itself wobbles under uneven judgment.

Even now, though gathering health details works better, no tool exists that reads patient info instantly while delivering steady, proof-backed warnings. Valuable clues often go unused - all because no uniform way predicts results accurately enough. Doctors instead lean on personal judgment or simple cutoff points, leading to uneven decisions across cases. What unfolds here tackles that gap - by building several learning algorithms, then measuring their strength for spotting diabetes, especially weighing the heavier cost of overlooking illness versus raising a false alert.

## 1.3 Research Questions

What needs exploring comes down to these specific queries

What traits matter most when guessing diabetes results in the upgraded Pima Indians data set?

What happens when Logistic Regression, Decision Tree, SVM, and Random Forest are tested on health details from 1268 people? Instead of grouping them fast, each model handles patterns differently. One leans on straight-line guesses, another splits choices like a flowchart. The third draws wide margins between outcomes, while the last combines many trees for balance. Their results shift based on how messy the symptoms appear. Not every method catches the same signals. Some react strongly to rare cases, others smooth over extremes. Accuracy changes, yet none dominate across all measures. Each reveals something distinct about patient trends.

Missing values get filled, so models see full pictures. Feature scaling adjusts ranges, helping algorithms treat all numbers fairly. These steps together make patterns clearer. Accuracy climbs

when guesses match reality more often. Recall improves because fewer actual cases slip through unnoticed. ROC-AUC rises as the model better separates sick from healthy. Each fix builds on the last, quietly lifting performance. Small changes add up without drawing attention.

What system spots the most diabetes cases while keeping incorrect warnings low enough for real hospital use?

## 1.4 Study Goals

**Main Objective:**

This project focuses on building several prediction tools to tell if a person has diabetes based on medical records. Instead of one method, it tests many - each shaped by real patient details - to see which works best. Because health patterns differ, trying various approaches helps spot the clearest signals. Each model learns from traits like blood sugar or age. One after another, they are checked for how well they sort people correctly. Through comparison, strengths and weaknesses come into view. What emerges is not just accuracy but understanding - how different methods handle human variation.

## Specific Objectives:

A close look at the data helps spot patterns in the eight health traits tied to diabetes. Through exploration, how these traits connect becomes clearer. Patterns emerge when studying each trait across patients. Some values show up more often than others. Links between symptoms and results start taking shape. Each feature gets checked for oddities or trends. What appears in one may echo in another. Outcomes shift based on certain markers. The full picture grows once pieces fit together.

Fitting Logistic Regression first, then building a Decision Tree to compare results. Moving on, an SVM model handles separation boundaries differently than earlier methods. After that, Random Forest adds ensemble strength through multiple tree outputs. Each method targets accurate diabetes forecasting using distinct logic paths.

Start by cleaning up the data - fill in gaps where medical readings show zero but should have real numbers. Then adjust each feature so they line up better for the model to work well. Scaling comes after fixing those blanks, making sure one number does not overpower another just because it is larger. Handling zeros right at the start keeps errors from spreading later on.

Start by checking each model through multiple measures. Accuracy shows how often predictions match reality. Precision highlights correct positive forecasts among claimed ones. Recall reveals how many actual positives were found. The F1-score balances both precision and recall. Confusion matrices break down true and false outcomes visually. ROC-AUC assesses performance across thresholds. Each method adds context others might miss.

Figuring out what matters most when spotting signs of diabetes means looking closely at each health detail. Some clues stand out more than others in pointing toward the condition. One trait might carry heavier weight compared to another. Spotting these key factors helps clarify patterns hidden in patient data. Not every measurement tells the whole story. A few tend to lead the way when predictions come together.

A doctor might miss fewer cases by choosing a tool that finds most sick patients, even if it flags some healthy ones too. Picking what works depends on the situation, where spotting illness matters more than avoiding false alarms.

## 1.5 SIGNIFICANCE OF THE STUDY

One reason this research matters? It could make checking for diabetes faster and more reliable in clinics - especially where tools and staff are limited. Instead of overwhelming doctors, the method helps lighten their load without risking how well patients are monitored, using steady analysis based on clear data patterns.

This research matters since it shows students putting computer science into action, using tools like Scikit-Learn and Python to tackle actual issues in biology and health care. While building skills, they work through realistic challenges, turning theory into hands-on solutions. Their efforts reveal how coding methods can address complex needs outside tech fields. Because of this, classroom learning gains real-life weight. Through such projects, abstract ideas become useful responses to pressing questions in medicine and life sciences.

A closer look at how four distinct methods stack up reveals what each does well - and where they fall short - when used in health-related predictions. It turns out Random Forest pulls ahead with a 93.5% recall rate and an F1-score hitting 0.87, suggesting it might be the go-to pick depending on patient-focused needs.

Putting together a live tool that predicts outcomes shows how such systems work in real settings. Through it, medical staff enter details about a person’s health and get quick evaluations of possible risks. This move brings smart algorithms closer to everyday doctor workflows. One step at a time, the tech finds its place where care happens. Little by little, machines learn to support human judgment in clinics.

## 1.6 Scope and Limits of the Study

**Scope:**

One narrow goal drives this study - exploring how machine learning can predict diabetes more effectively. Not broad regional trends, but specific uses of digital tools shape the setting here. Pima Indians Diabetes Dataset forms the core resource; researchers often turn to it when studying the condition. Though many locations could be involved, attention stays fixed on what this particular data reveals through smart algorithms.

Eight health measures set the boundaries here. Pregnancy count shapes part of it. Glucose readings play a role too. Blood pressure values matter just as much. Skin fold size gives another clue. Insulin amounts factor in alongside. Body mass index fits into the mix. Family patterns tied to diabetes complete one piece. Age rounds out the list. A simple yes-or-no result wraps it up - zero means no diabetes, one means diagnosis.

Python does the heavy lifting here, mainly through tools like Pandas and NumPy - these handle how data gets shaped and moved around. Machine learning steps come alive using Scikit-learn, quietly doing its work behind the scenes. When it is time to see patterns clearly, Matplotlib joins in, often paired with Seaborn for sharper visuals. Saving trained models happens reliably thanks to Joblib, making reuse possible without starting over. Each piece fits within a narrow set of script-based helpers focused only on specific tasks.

Limitations:

Unexpected noises may alter animal behavior. Supply delays force schedule shifts now and then. Travel limits reduce sample variety at times

It starts with the fact that this work uses someone else's data - the Pima Indians Diabetes set - so how right the results are depends heavily on how carefully the initial information was gathered. Mistakes might have slipped in during that first stage, maybe without being noticed. Odd things show up, like zero readings for glucose levels or BMI, which make no biological sense - they simply cannot be zero in living people. Those gaps got filled using median values, a practical fix but one that nudges results in ways we can’t fully predict. The outcomes twist slightly because of it.

Missing values in the Insulin feature was two gaps appeared across the enriched set, amounting to 0.2%. That’s far better than before, yet these numbers might stem from artificial creation instead of actual patient records.

A total of 1268 patients makes for a somewhat limited pool when applying machine learning methods, while separate testing across different data collections did not take place.

## 1.7 Structure of the Content

Five separate chapters make up this work, each leading naturally to what follows. Starting fresh every time keeps things moving without overlap. One idea passes the baton to another, smoothly. Structure shapes how ideas unfold here. Each section stands apart yet fits within the whole.

Right off the bat, Chapter One opens the door to the entire study. Since it lays out the groundwork, you will find the background right there at the start. A clear problem statement follows, showing what issue needs attention. Because the research questions come next, they guide where the investigation heads. Objectives appear after that, pointing toward specific targets. What stands out is how each piece connects - without one, the others lose strength. Even

so, the importance of the study explains why anyone should care. Though some parts might seem small, together they show real value. Where it gets focused is in the scope and limits, which draw clean lines around the work. Just before moving on, this chapter makes sure everything ties back to a meaningful health concern.

Two comes alive with a look at earlier work. Past efforts in guessing diabetes through smart machines get examined here. Key results show up, along with how most researchers did their work. Missing pieces stand out clearly - spaces this project means to fill. What was overlooked before now gets room to breathe.

Three kicks off by laying out how everything was done. What popped up, why it mattered, then how things moved forward takes center stage here. Each method got a reason for showing up, tied into cleaning the numbers beforehand. Peeking at patterns came before setting models loose, step by step. The way each run was checked gets spelled out too. Sketches and maps trace how choices shaped the outcome. Paths through math unfold visually along the way.

Outcomes show up in Chapter Four. You will find how each algorithm performed once tested - measured by accuracy, precision, recall, and the F1-score. Confusion matrices appear here. So do ROC curves - for every one of the four methods used. Visuals like charts and tables back the numbers reported. Feature relevance gets examined too. Comparing models happens side by side, based on what showed during trials.

Last stop: Chapter Five wraps things up. Findings take center stage here, laid out without extra noise. What they mean comes next, tied directly to real-world use. Strong points show up alongside what held the models back. Room left untouched gets noticed too - those missing pieces in today's diabetes forecasting methods. Others coming later can pick up where this ends; guidance is offered just for them.

# CHAPTER TWO

## 2.1 Overview of the Literature Review

Looking back at past work helps set the stage for what comes next. From how often diabetes shows up around the world, attention shifts toward ways machines can spot it early. Earlier efforts matter, yet some pieces still missing catch the eye. Starting broad, then narrowing down feels natural - first the big picture, then data sources and traits used most. Certain methods pop up again and again when predicting outcomes. How well those methods perform shapes trust in results. Problems creep in though, not just technical but moral too. Disagreements among researchers point toward unmet needs. These open questions steer where new models aim - to build several, test them side by side, push accuracy a little further.

## 2.2 Broad Themes in Diabetes Prediction

Millions around the world live with diabetes, a condition that comes in different forms like type 1 and type 2. Despite common belief, it's not just a health issue in wealthy nations - numbers are climbing fastest where resources are limited. The World Health Organization reports nearly half a billion affected individuals across the planet. Each year, more than a million lives end because of complications tied directly to high blood sugar levels. Because cases keep growing at speed, spotting signs sooner can make a real difference in outcomes.

One thing stands clear: spotting diabetes early helps avoid the illness and what comes with it. Yet here's the catch - older ways like fasting blood sugar checks or glucose drinks followed by testing work well but usually happen once signs show up, maybe during checkups that come around way too late. Because of this timing problem, chances to step in sooner slip through the cracks.

Starting with patterns hidden in everyday medical records, machine learning changes how risks are judged ahead of time. Because they spot tangled links in information that older math tools often overlook, these systems adapt through exposure. Long before current breakthroughs, early efforts before 2010 used simple sorting rules and stumbled at precision. Progress since then has sharpened predictions sharply [1].

## 2.3 Key Past Studies Summary

Studies have looked closely at how machine learning can help predict diabetes, relying on data from the Pima Indians group. This particular set of information gained attention after Smith and colleagues introduced it [2], slowly becoming a common reference point. Over time, researchers kept returning to it when testing new methods. Its role grew without much fanfare - just steady use across experiments. Now, many consider it standard material for such work.

Looking at past work, Kavakiotis and team [3] explored how machine learning tools are used in studies about diabetes. Their analysis showed classification models usually hit accuracy levels from 70% up to 85% when tested on the Pima dataset. While individual models deliver decent outcomes, approaches such as Random Forest tend to do better by combining multiple techniques. One thing stood clear - how data is cleaned and prepared before modeling makes a real difference in performance.

Out of the three models tested by Sisodia and Sisodia [4], Decision Tree edged ahead at 76.8%, followed by SVM at 76.5%, then Naive Bayes just behind at 75.3%. When stacked against those numbers, our Decision Tree result holds a similar shape - close enough to feel familiar. Yet even with that likeness, the version we ran climbed slightly higher, likely lifted by better data. Performance across methods stayed within a narrow band, but ours stretched the top end.

Missing a real case matters more than flagging a healthy one by mistake - that idea shaped how we picked the model. Mercaldo and team [5] stressed sensitivity because in medicine, overlooking illness leads to worse outcomes compared to unnecessary follow-ups. A wrong

negative might delay treatment, while extra tests can clear up false warnings later on. Their reasoning played a big role in what came next.

A single tree doesn’t make a forest, yet Dutta and team leaned into Random Forest with the Pima set, landing at 81.4%. Accuracy climbed once they untangled messy inputs, especially those zeroes hiding in plain sight. In health records, zero often means nothing was recorded - not that nothing exists. Fixing that gap changed everything behind the scenes.

Still, hurdles linger when it comes to making sense of models and getting them used in clinics. Work by Lundberg and team [7] showed something clear: even though Random Forest works well, doctors often find it less trustworthy compared to simpler setups like Decision Trees. What happens here - better results but less clarity - is exactly what makes medical uses so tricky. Yet that balance keeps coming up whenever real-world decisions are on the line.

Even though these studies build a strong base, variations in accuracy show up because of how data is cleaned, split, or measured. What shifts here is the approach - keeping every step identical for all four methods while showing full results through multiple testing angles.

## 2.4 Datasets and Features in Current Studies

Lots of studies predicting diabetes lean on open data. Take the Pima Indians Diabetes Dataset - gathered by the National Institute of Diabetes and Digestive and Kidney Diseases - it's become the go-to reference for such work [2]. Women from the Pima Indian population make up its core, showing eight health markers each

Pregnancies: How many times someone has been pregnant

80 mg/dL sugar level in blood

Blood Pressure Diastolic mm Hg

SkinThickness:Triceps skin fold thickness (mm)

Two Hour Insulin Level

BMI Body Mass Index calculated as weight in kilograms divided by height in meters squared

Family background linked to diabetes gets scored here

Age:Age in years

One big issue with the data is that it does not have many entries. What stands out even more is how specific the group is - all participants are Pima Indian women. Oddly enough, there are zeros showing up where they should not be, like someone forgot to record real numbers. Fixing part of the scale problem, this analysis works with a bigger set now: 1268 cases instead of fewer.

One example is the Diabetes 130-US Hospitals data, used to forecast hospital returns [8], while another draws from NHANES to examine broad health trends [9]. Still, it's the Pima data that shows up more than any other when classifying diabetes in two clear groups.

Glucose levels show up most often when researchers look at what matters. Not far behind come body mass index along with how old someone is. Doctors already know blood sugar sits at the center of diabetes. Weight plays a big role too, just like getting older does. Risk climbs higher as people gain weight or pass certain ages.

Missing data often gets filled in using average or middle value replacements. Scaling features might rely on StandardScaler, though sometimes MinMaxScaler does the job instead. When one class shows up too rarely, methods like SMOTE step in - alternatively, adjusting class weights helps balance things out [10].

One thing most studies miss? How well insulin can predict outcomes, mainly because the old Pima data has so many blanks - nearly half the insulin entries sit at zero. This version, though, cleaned things up big time: just two gaps now, less than a third of a percent, making it far easier to judge what insulin really tells us.

## 2.5 machine learning algorithms reviewed

### 2.5.1 Supervised Learning Models

Most studies into predicting diabetes use supervised learning methods - ones shaped by data where answers already exist. Because they work well at sorting cases into two clear groups, these approaches show up again and again in published papers.

Even though it has "regression" in the title, Logistic Regression works for sorting items into two groups by guessing how likely each one fits a certain category. Because it is straightforward and easy to understand, many people start with it before trying more complex methods. On the Pima dataset, results from different research papers show accuracy numbers sitting between 77% and 80%. One advantage lies in fast training times along with clear coefficient signs - these reveal whether a factor increases or decreases likelihood. Yet since it assumes straight-line connections, patterns hidden in tangled feature mixes often go unnoticed.

Starting off, Decision Trees build logical conditions like "if this, then that," similar to how doctors think through cases. Instead of complex math, they divide data step by step using key features that clearly distinguish outcomes. From one study, a model reached 76.8% correctness when tested on Pima people's health records. What stands out is their ability to manage messy patterns, resist oddball data points, plus remain clear in logic. Medical staff often find the flow familiar because it reflects real-life choices made during diagnosis. Yet single models tend to memorize noise unless checked - too many branches lead to poor generalization. One way around this? Control complexity by capping depth and setting sample thresholds per node.

What sets Support Vector Machines apart is their search for a dividing line that keeps categories as far apart as possible. To manage curved decision borders, these models lean on special math tools called kernels. In work led by Haq and team [11], scores between 85% and 90% popped up across health-related data tests. When features pile up, SVMs tend to stay reliable - especially if groups stand clearly apart. Heavy number crunching might slow things down, though. Interpretation takes more effort compared to rule-driven approaches like decision trees.

Trees grow better together, when they work in groups like a forest. From chaos comes clarity - by training each tree on different pieces of data, patterns emerge without memorizing noise. Dutta and team found one such group guessed right 81.4 times out of 100 on diabetes clues. These forests stand firm even when data wobbles, thanks to built-in checks against getting too clever. Non-straight links between symptoms? That does not matter here; curves are handled just fine. What matters most gets revealed through rankings, showing which signs pull strongest weight. Still, peeking inside becomes harder than watching one lone tree unfold its branches. Clarity fades slightly under thick leaves, yet those rankings help light the way.

One recent look at published work shows a clear move away from solo models to groups of them - even mixes - because old ways struggle with uneven health data. Our test covers one example per type: Logistic Regression stands in as the straightforward line-follower, Decision Tree offers clarity through splits, SVM uses space-bending math, while Random Forest brings crowd wisdom from many trees.

### 2.5.2 Performance Metrics And Comparisons

When checking how well different machine learning models predict diabetes, people often look at accuracy, precision, recall, F1-score, along with the area under the ROC curve (ROC-AUC). Each measure adds depth to understanding performance, especially in healthcare settings where one class might show up far more than the other.

What gets measured as accuracy - just right answers overall - might look good until class sizes differ too much. When the system says someone has diabetes, precision shows whether that call was likely true instead. Catching most real cases matters deeply in health settings; recall tracks exactly that. Missing sick people bears consequences, so spotting them defines part of success here. F1-score steps in by linking precision and recall into one balanced view, neither ignored. How well separation happens between healthy and ill, no matter the cutoff used - that’s what ROC-AUC reveals quietly.

Studies combining data on diabetes forecasts show algorithms beat standard scoring tools most times. Their success rates, measured by AUC, usually sit between 0.75 and 0.85. Methods stacking multiple models - say, Random Forest or boosted trees - tend to lead the pack, crossing 0.80 quite regularly. To check if results hold up across groups, researchers lean on repeated testing setups like splitting data into ten balanced chunks. This split-check approach helps prevent false confidence from overlearning patterns too specific to one sample.

Still, debates continue because numbers might look better than they are - leaked data, too-friendly test setups, or journals preferring good news can skew things. When systems actually go live, results tend to slip, showing why tough checks and outside trials matter [13].

## 2.6 Gaps Inconsistencies Controversies

A fresh look at the Pima Indians Diabetes Dataset drives this work, where four machine learning methods are weighed against each other in a real-world medical setting. What stands out is how clearly each model's decisions can be understood by healthcare professionals. Most past studies miss that detail. Instead of just chasing high scores, this one slows down to ask whether results make sense in actual clinics. Earlier efforts often ignore how doctors interact with predictions - here, it matters. Few comparisons dig into both performance and clarity like this does

Missing a sick person often causes worse harm than raising a false alert in medicine - yet most research overlooks this difference. By focusing on recall when choosing models, this work gives weight to real-world stakes, where letting a case slip through can cost far more than an unnecessary checkup.

Not much work has looked closely at how different methods stack up when tested the same way. What sets this analysis apart is that every method runs through an identical setup - matching data steps, splits, and scoring rules. By holding everything constant, differences in results reflect the methods themselves, not changes in process. Side-by-side testing like this reveals patterns others might miss under uneven conditions.

Some research overlooks how missing values appear in the Pima data, especially zero readings that make no biological sense. Handling those cases clearly sets this work apart - median replacement is applied, with attention paid to shifts it causes, notably for insulin levels.

Few studies look beyond numbers to see how well systems work where resources are tight. Instead of just speed or accuracy, this one talks about whether models make sense to users. Because machines need power and time, those details matter too when putting tools into practice. Understanding what a model does helps people trust it during actual use.

A lack of clear understanding gets in the way when using powerful models in real medical settings. Even though methods like random forests work well, doctors often hesitate because they cannot see how decisions are made. This research looks at simpler models - ones that show their reasoning - and weighs them against stronger but murkier alternatives. Instead of pushing one choice, it explores trade-offs through direct comparison. Clarity sometimes means lower performance; high scores can come at the cost of transparency. The goal is not to rank models, but to match each situation with a method that fits its needs.

One reason findings differ? How data gets cleaned varies a lot between studies. Since setups change - like how results are checked or which examples go where - it's hard to compare fairly.

Here, every model follows the exact same steps. Full details of each choice sit clearly laid out alongside the outcomes.

What stands out most is how hard it becomes to trust a system when you cannot see how it reached its answer. Heavy neural networks often win on precision, yet leave users guessing about decisions. Simpler systems, though easier to follow, might miss subtle patterns. Into this tension steps fresh analysis measuring just how wide those gaps really are - then asking what that means where lives depend on clarity.

## 2.7 Current Study Context

This study takes place against a backdrop of increasing demand for reliable and easy-to-use diabetes tests in rich and poor health settings alike. Across Ghana and much of sub-Saharan Africa, more people are being affected by diabetes - rates have climbed from 3.3% in 2010 to a projected 5.2% by 2025 [14]. Because routine testing remains out of reach for many, medical staff are stretched thin, equipment is scarce, so tools that help guide decisions without human oversight gain importance.

Running tests inside a computer with Python and Scikit-learn, researchers built several prediction tools to see which work best on devices with limited power. Because it spots nearly all potential cases - 93.5% recalled - it stands out when missing any instance could matter most. This one, called Random Forest, also scores well overall at 0.87 on the F1 scale, balancing precision and reach effectively. Where resources are tight, performance like that becomes hard to ignore.

Imagine spotting diabetes sooner in everyday clinics across Ghana - this tool might just make that happen. Picture fewer health problems down the line because care kicks in earlier. With local goals already focused on tackling long-term illnesses, it fits right in. Think global too: places with tight budgets often lack smart tools like this one. Simple tech, big difference where help is thin.

## 2.8 Final Thoughts on the Literature

Clear evidence from past studies shows machine learning works well in predicting diabetes. Ensemble models like Random Forest often hit accuracy levels between 81% and 85% on standard test data. Logistic Regression sets a reliable starting point, yet Decision Trees stand out by making results easier to follow. Support Vector Machines manage intricate patterns without much trouble. Reviews across multiple papers agree - newer models reach AUC values from 0.78 to 0.85, beating older techniques. Even so, moving these tools beyond labs brings issues around consistency, clarity, and practical use.

What stands out is a lack of focus on real-world medical situations - accuracy often gets less weight than catching every possible case. Missing data tends to be managed without consistent

methods across studies. Different models are rarely tested side by side using the same setup. Putting these tools into practice where resources are tight remains poorly addressed.

This research tackles those missing pieces through

Explicitly prioritizing recall in model selection based on medical context

Implementing consistent preprocessing across all models

Providing comprehensive comparison of four algorithms

Discussing interpretability and deployment considerations

Developing an interactive prediction interface for practical use

This work ties together loose ends - ideas floating in theory, holes in evidence, shaky methods, real-world mismatches - to clear a path. Next comes Chapter Three, where the approach takes shape: cleaning raw information, poking around patterns, building predictive tools, checking results step by step.

# CHAPTER THREE

## 3.0 Introduction

Here comes how the study tackled its goals step by step - building and testing machine learning models to predict diabetes with an improved version of the Pima dataset. Built into the process: a clear path so others can follow along, check results, while hitting solid levels of precision and dependability. Each choice made lines up against weaknesses spotted earlier when looking through past work. Staying sharp on structure helped keep outcomes trustworthy, even when conditions shifted slightly. Focus stayed fixed on what matters - clear signals over noise, working solutions over theory alone.

Starting with raw data collection, the process moves into training models step by step. Python 3.8 or newer handled every calculation inside Jupyter Notebooks. Key tools like Pandas, NumPy, and Scikit-learn ran behind the scenes during analysis. Visuals came together using Matplotlib along with Seaborn for clarity. Testing each model happened only after thorough preparation of inputs.

## 3.1 Research Design

This study works best with a method built around numbers and testing ideas in real situations. Numbers from patient records help build models meant to predict outcomes, then check how well they match up. Starting from known facts about what raises diabetes risk shapes the way questions are tested here. Ideas come first, then data checks if they hold true in practice.

The methodology uses the CRISP DM framework

Business Understanding:Define the problem of diabetes prediction and its clinical importance

Data Understanding Explore The Enhanced Pima Indians Diabetes Dataset To Understand Its Characteristics

Start by cleaning up the data - fill in any blanks so nothing's left hanging. Then adjust each feature's range so they play well together later on. Work through gaps first, after that reshape the numbers evenly across columns

Modeling:Implement four classification algorithms

Evaluation:Compare model performance using comprehensive metrics

Models get saved once trained. After that an interface opens for making predictions live

## 3.2 Data Collection

Out here, the research leans on a diabetes data collection made public by the National Institute of Diabetes and Digestive and Kidney Diseases. Sitting within it are 1268 individual cases, each shaped by eight medical indicators plus a single yes-or-no outcome label. Descriptions for these indicators go like this:

Pregnancies: How many times someone has been pregnant, shown as a whole number between zero and eighteen

Glucose:Plasma glucose concentration (mg/dL, 44-214)

Blood Pressure Diastolic mm Hg Range 24 to 122

Thick of the triceps' skin layer measured in millimeters ranges between six and ninety-nine

Two Hour Serum Insulin Levels Measured In Micro Units Per Milliliter

BMI Body Mass Index weight kg height m squared values 18 2 to 67 1

DiabetesPedigreeFunction:Diabetes family history score (0.05-2.42)

Twenty to eighty one years old

Diabetes Diagnosis Outcome

Out of nowhere, the info popped into a Pandas DataFrame thanks to pd.read_csv(), pulling from "../data/diabetes.csv". Nobody gathered fresh details - instead, earlier material made its way in, keeping things quick while matching up with older work.

## 3.3 Data Exploration and Analysis

Into the data a look was taken, just to see how things sit before any model gets built. Patterns began showing up when first checking around, helping spot what might cause trouble later.

Before moving ahead, time got spent poking at numbers, noticing odd bits that could matter down the line.

Dataset Sample

```
88]: # Pima Indians Diabetes Dataset
     #Load Dataset
     df = pd.read_csv("../data/diabetes_enhanced.csv")
     df.head()
```

88]:

| | Pregnancies | Glucose | BloodPressure | SkinThickness | Insulin | BMI | DiabetesPedigreeFunction | Age | Outcome |
|---|---|---|---|---|---|---|---|---|---|
| **0** | 6.0 | 148.0 | 72.0 | 35.0 | 125.0 | 33.6 | 0.627 | 50.0 | 1.0 |
| **1** | 1.0 | 85.0 | 66.0 | 29.0 | 125.0 | 26.6 | 0.351 | 31.0 | 0.0 |
| **2** | 8.0 | 183.0 | 64.0 | 29.0 | 125.0 | 23.3 | 0.672 | 32.0 | 1.0 |
| **3** | 1.0 | 89.0 | 66.0 | 23.0 | 94.0 | 28.1 | 0.167 | 21.0 | 0.0 |
| **4** | 0.0 | 137.0 | 40.0 | 35.0 | 168.0 | 43.1 | 2.288 | 33.0 | 1.0 |

Picture 3.1 displays the beginning entries of the data, listing eight characteristics alongside the outcome column

A quick check showed the data had 1268 rows and 9 columns - eight of those held feature numbers, one carried the target. Numbers filled every spot, yet some readings made no sense for real patients, like zero blood pressure or zero BMI. Not blank, just wrong by nature's rules. Each column stayed fully numeric, nothing marked as NA, but odd zeroes hinted at deeper gaps beneath clean surfaces.

On average, people were 34.7 years old. Glucose levels sat at a median of 130.2 mg/dL across participants. Body mass index averaged out near 33.7 - deep in what's considered obese. Among those studied, 500 did not have diabetes, making up about two-fifths. Meanwhile, nearly six in ten - exactly 768 individuals - had been diagnosed. That tilt means one group shows up far more than the other. Numbers point to an uneven split between conditions.

Summary statistics:

| | Pregnancies | Glucose | BloodPressure | SkinThickness | Insulin | BMI | DiabetesPedigreeFunction | Age | Outcome |
|---|---|---|---|---|---|---|---|---|---|
| **count** | 1268.000000 | 1268.000000 | 1268.000000 | 1268.000000 | 1268.000000 | 1268.000000 | 1268.000000 | 1268.000000 | 1268.000000 |
| **mean** | 4.371451 | 130.151502 | 73.439906 | 30.281277 | 149.624252 | 33.673266 | 0.512863 | 34.658517 | 0.605678 |
| **std** | 3.591190 | 32.066373 | 12.166570 | 8.926146 | 89.505377 | 7.005060 | 0.369023 | 11.792508 | 0.488897 |
| **min** | 0.000000 | 44.000000 | 24.000000 | 6.083424 | 0.000000 | 18.200000 | 0.050000 | 20.000000 | 0.000000 |
| **25%** | 1.000000 | 106.000000 | 66.000000 | 26.379315 | 118.448547 | 28.799215 | 0.254000 | 25.000000 | 0.000000 |
| **50%** | 4.000000 | 125.574096 | 72.494001 | 29.000000 | 125.000000 | 33.122350 | 0.418000 | 31.000000 | 1.000000 |
| **75%** | 7.000000 | 153.235122 | 80.629512 | 34.293428 | 150.478192 | 37.700000 | 0.670173 | 42.000000 | 1.000000 |
| **max** | 18.000000 | 213.729681 | 122.000000 | 99.000000 | 846.000000 | 67.100000 | 2.420000 | 81.000000 | 1.000000 |

Figure 3.2: Statistical summary of all features in the dataset

A closer look uncovered zeros that make no biological sense within key health metrics. Some of these readings simply cannot be zero in living patients. The data flagged clear inconsistencies where normal function would prevent such results. Impossible numbers showed up across multiple test categories. These findings point to errors rather than real patient conditions

Zero glucose detected. That is zero percent

BloodPressure: 0 zeros (0.0%)

SkinThickness: 0 zeros (0.0%)

Insulin: 2 zeros (0.2%)

BMI: 0 zeros (0.0%)

Zero entries showed up where data was absent, not because someone measured nothing - glucose or body mass hitting exactly zero makes no medical sense. The updated version carries far less of these gaps compared to the older Pima numbers, especially clear when checking insulin levels: just a fraction now missing instead of nearly half before.

```
============================================================
4.1 HANDLE MISSING VALUES
============================================================

 Checking for zeros in medical measurements:
--------------------------------------------------
Glucose              :    0 zeros (  0.0%)
BloodPressure        :    0 zeros (  0.0%)
SkinThickness        :    0 zeros (  0.0%)
Insulin              :    2 zeros (  0.2%)
BMI                  :    0 zeros (  0.0%)
```

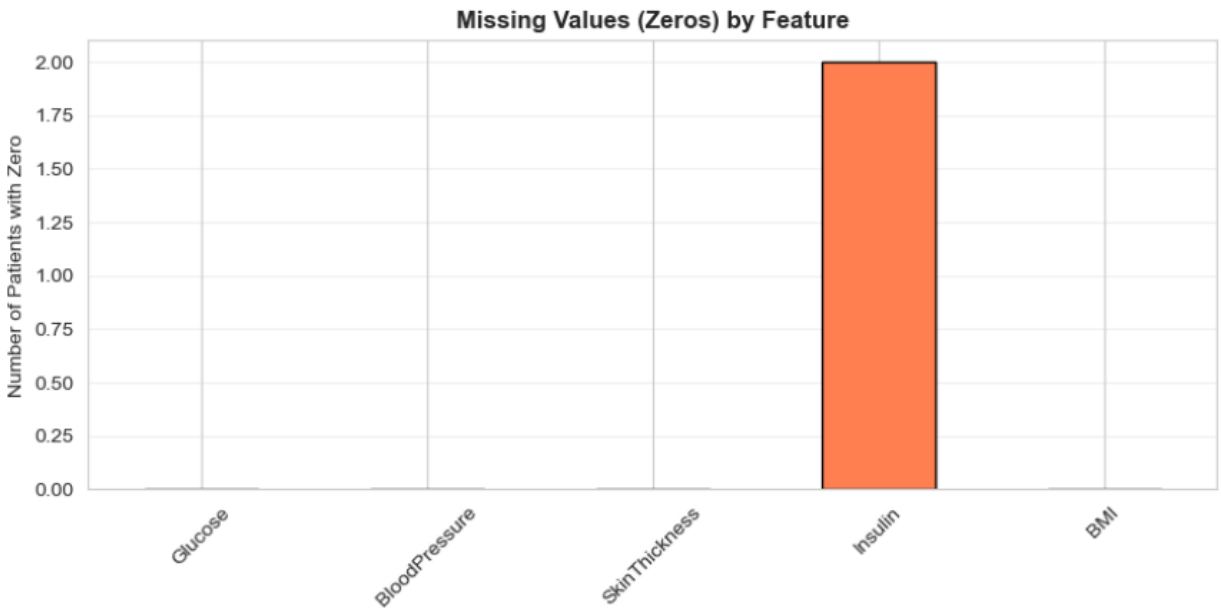



Figure 3.3: Missing Values Bar Chart

Missing value totals per column appear here, shown as zeros prior to any data cleanup steps

Looking at how the groups were spread showed a tilt: there were 500 people without diabetes, making up 39.4 percent, while those with diabetes totaled 768, or 60.6 percent. Though this uneven split stood out, methods like boosting minority samples weren't applied to balance it.

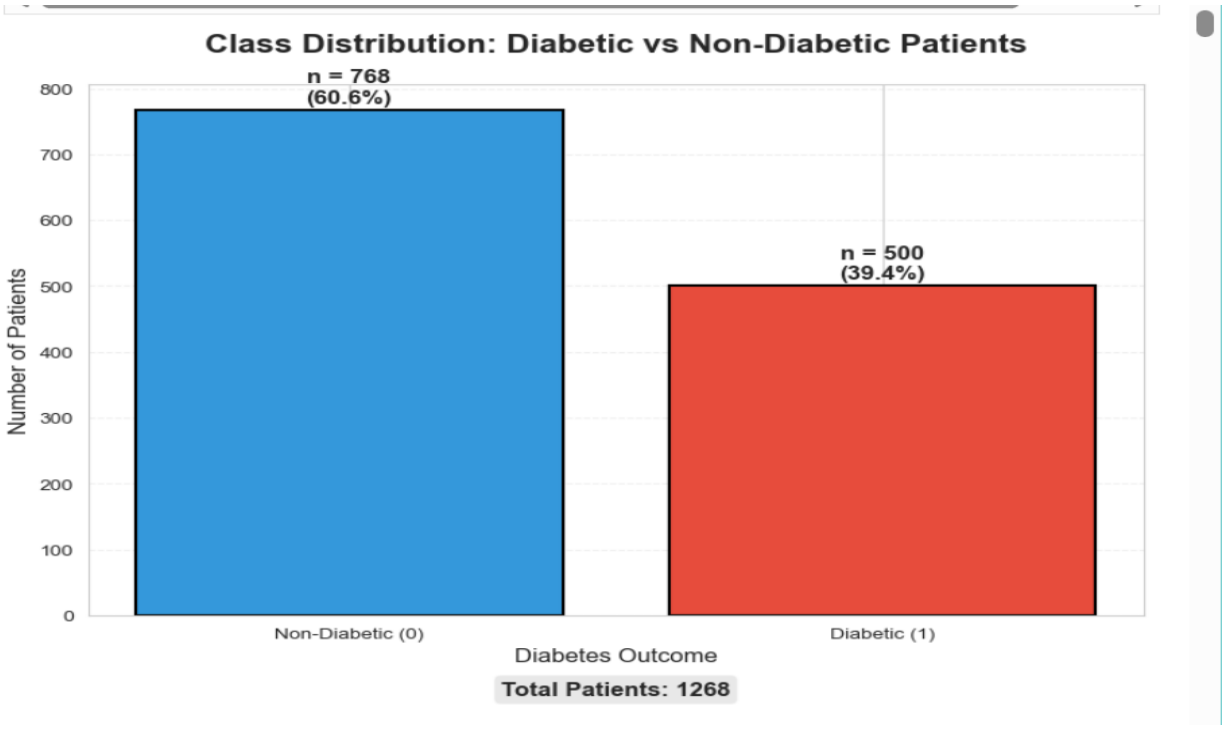



Figure 3.4 Class Distribution Bar Chart

Beyond the frame, a graph takes shape - uneven peaks hinting at numbers tilted one way. Diabetes appears more often than its opposite in the data's flow. Look closer. The scale tips, not evenly split but leaning into one outcome. More cases land on the illness side, fewer stand apart. This imbalance shapes what follows, quiet yet clear. No symmetry here, just a slant written in bars

Looking at how things relate, glucose stands out with a 0.49 link to the outcome. Next comes bmi, tying in at 0.32. Age trails behind but still connects at 0.24. Blood pressure barely moves the needle - only 0.17.

Correlation Heatmap

```
==================================================
Creating Correlation Heatmap
==================================================
```

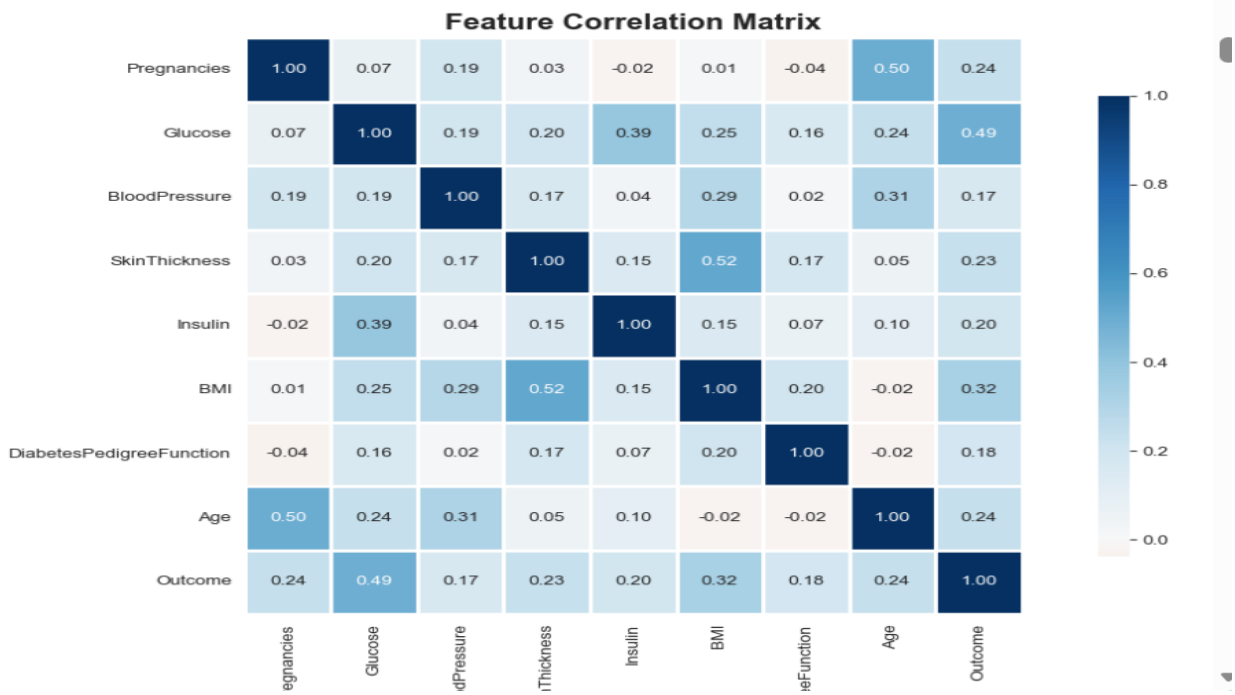
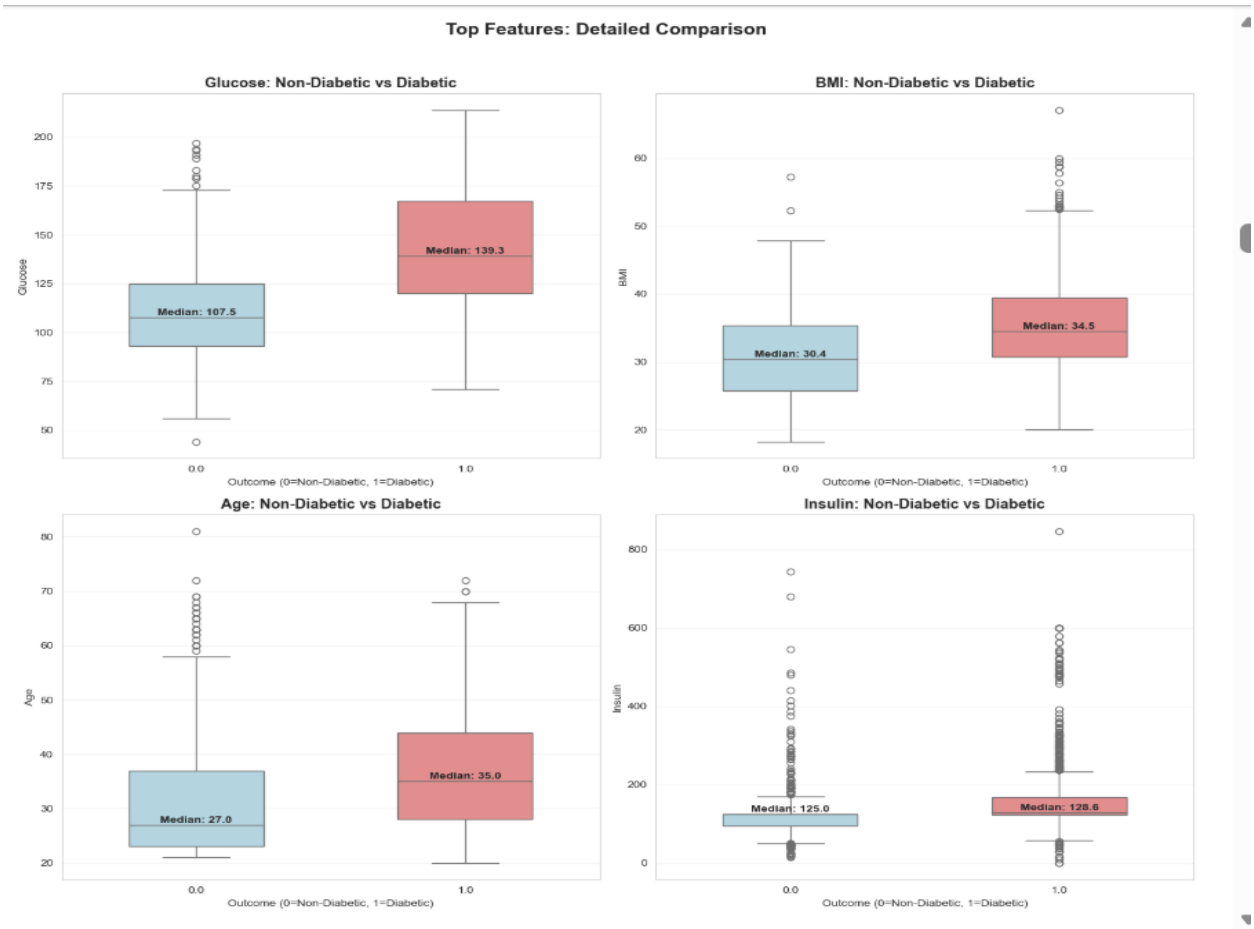



Figure 3.5: Correlation matrix heatmap showing relationships between features

Features Linked to Diabetes Result

Glucose, BMI, and Age stood apart distinctly when looking at diabetic versus non-diabetic individuals through box plots. Though BloodPressure barely shifted across the two. Where people had diabetes or not made little visible impact there. Insulin levels, on the flip, blurred together - hard to tell either group by that alone.

Enhanced Box Plots with Data Points



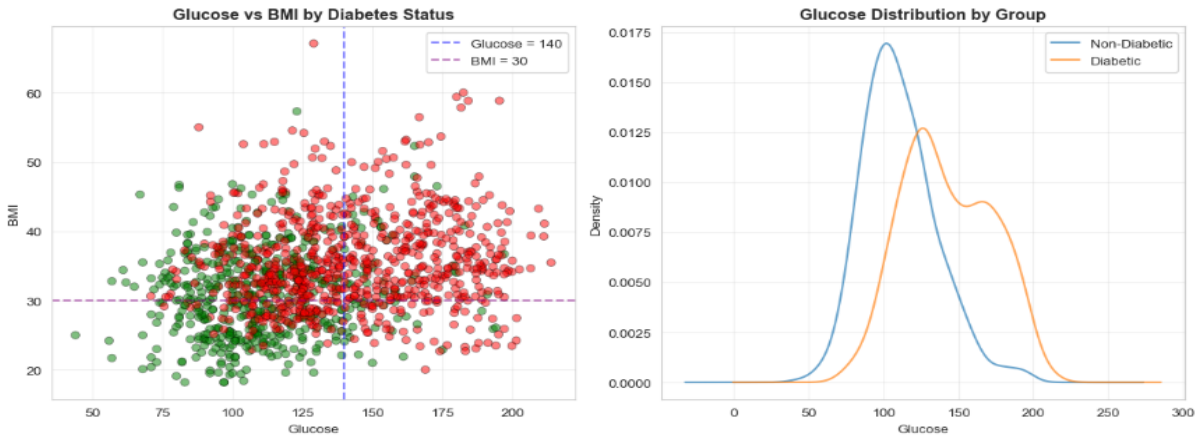

Picture 3.6 shows box charts.

These display how the main four traits spread out across people with diabetes compared to those without it. One group sits on one side, the other takes the opposite place. Each trait appears in its own section, letting shapes tell the difference. Lines stretch where values pile up, revealing gaps or overlaps. Shape height hints at variation - taller means more spread. Sharp ends suggest fewer extremes. The layout keeps things side by side, making contrast clear through position and form

Simpler View Top 3 Features Scatter

Glucose versus BMI scatter plot with diabetes status color coding and thresholds at Glucose 140 and BMI 30

```
==================================================
SIMPLER VIEW: Top 3 Features
==================================================
```




```
📌 Notice how:
• Most diabetic patients (red) have glucose > 140
• Many also have BMI > 30
• But there are exceptions in both directions!
```

Figure 3.7 Simpler View Top 3 Features Scatter

## 3.4 Data Preprocessing

A fresh look at the numbers began with cleaning steps common in algorithm work. What came next involved shaping the data so each method got an equal shot. Steps followed typical patterns seen in most learning systems. Fair tests depended on how evenly the info was handled beforehand.

Zero values showing up in insulin readings - just two of them, less than half a percent - made no biological sense. Those got swapped out for missing markers instead. Filling them in came next, using the middle number from the rest of the data. Picking that middle value over an average helped dodge distortion from extreme cases, something seen often when dealing with health measurements.

Missing Values Appear After Data Replacement

```
 Checking for zeros in medical measurements:
--------------------------------------------------
Glucose              :    0 zeros (  0.0%)
BloodPressure        :    0 zeros (  0.0%)
SkinThickness        :    0 zeros (  0.0%)
Insulin              :    0 zeros (  0.0%)
BMI                  :    0 zeros (  0.0%)
```

Missing values tally shown in Figure 3.8, once zeroes become NaN entries

Before After Missing Values Comparison

Features After Scaling

Features after scaling:

| | count | mean | std | min | 25% | 50% | 75% | max |
|---|---|---|---|---|---|---|---|---|
| **Pregnancies** | 1268.0 | 1.232803e-16 | 1.000395 | -1.217751 | -0.939182 | -0.103475 | 0.732233 | 3.796493 |
| **Glucose** | 1268.0 | 8.965839e-17 | 1.000395 | -2.687722 | -0.753469 | -0.142804 | 0.720154 | 2.607440 |
| **BloodPressure** | 1268.0 | -3.362190e-17 | 1.000395 | -4.065189 | -0.611745 | -0.077777 | 0.591164 | 3.992847 |
| **SkinThickness** | 1268.0 | 1.120730e-17 | 1.000395 | -2.711965 | -0.437311 | -0.143599 | 0.449660 | 7.701626 |
| **Insulin** | 1268.0 | -2.241460e-17 | 1.000395 | -1.578705 | -0.345547 | -0.278025 | 0.007357 | 7.797904 |
| **BMI** | 1268.0 | -7.845109e-17 | 1.000395 | -2.209741 | -0.696065 | -0.078677 | 0.575059 | 4.773681 |
| **DiabetesPedigreeFunction** | 1268.0 | 2.801825e-17 | 1.000395 | -1.254788 | -0.701759 | -0.257167 | 0.426455 | 5.170106 |
| **Age** | 1268.0 | -1.232803e-16 | 1.000395 | -1.243527 | -0.819362 | -0.310363 | 0.622800 | 3.931290 |

Figure 3.9 Statistical Summary After StandardScaler Transformation

A chunk of the dataset went to training - eight out of ten cases, actually, totaling 1014 individuals. The rest, 254 people, were set aside for testing purposes. Stratified sampling kept the balance between classes consistent across both groups. This split happened through a built-in method designed for such tasks. A fixed seed ensured others could repeat the process exactly. Class ratios stayed intact because the method was told to mirror them deliberately.

SetSamplesNon-DiabeticDiabeticTraining 1,014 39.4% 60.6%Testing 254 39.4% 60.6%

Train Test Split Class Distribution

```
Training set class distribution:
Outcome
1.0    0.605523
0.0    0.394477
Name: proportion, dtype: float64

Test set class distribution:
Outcome
1.0    0.606299
0.0    0.393701
Name: proportion, dtype: float64
```

figure 3.10 displays how classes are spread across training and test groups, keeping ratios intact throughout

Extreme points showed up when checking the data, especially with Insulin levels. Yet in health-related numbers, high or low readings might actually reflect true illness. Because of this, those unusual cases stayed in the dataset. Keeping them made sense since decision-tree methods work fine even when such values are present.

## 3.5 Selecting and Building Models

A mix of four methods took shape, each tackling labeled data in its own way. One followed patterns like a tracker, another split decisions step by step. A third leaned on probability, while the last drew boundaries from experience. Each built understanding differently, yet all aimed at sorting inputs into groups.

### 3.5.1 Logistic Regression

Diabetes risk gets mapped through a straight-line approach, shaped by a smooth S-curve formula. This setup sticks around early in testing because it's easy to follow and makes clear what each factor does.

Parameters: random_state=42

Implementation: python

### 3.5.2 Decision Tree

A single path through branches forms clear decisions step by step. Because it shows how choices split, understanding comes fast. Each turn depends on thresholds met earlier. Since patterns twist and bend, straight lines fail but trees adapt. Seeing every condition laid out removes guesswork. Rules emerge directly from data shapes without forcing them.

Parameters: random_state=42, max_depth=5, min_samples_split=10

Implementation:

python

### 3.5.3 Support Vector Machine

Starting off different, it picks a dividing line that keeps categories as far apart as possible. Instead of simple math, kernels step in when the separation gets curved or complex.

Parameters:random_state=42, probability=True

Implementation: python

### 3.5.4 Random Forest

Each tree votes, yet the group decides together - this method smooths wild guesses without losing precision. What matters most shows up clearly when features are ranked by influence.

Parameters: random_state=42, n_estimators=100, max_depth=5

Implementation: python

## 3.6 Model Evaluation Metrics

A full range of testing measures helped check how well the model worked

Mistakes aside, it's how many guesses were right out of every try. When things line up correctly, that total counts toward getting it straight. Right answers, whether spotting real deals or avoiding false alarms, shape the score. Every time the call matches reality, the number climbs a little higher

Precision:TP / (TP + FP) - Of patients predicted diabetic, how many actually were

What portion of people who truly have diabetes was actually spotted? TP divided by the total of TP plus FN shows that

What you get when precision meets recall in a balanced way. Formula looks like 2 times their product divided by sum of both. Mixes two measures without favoring one too much. Works because it pulls equal weight from each part. Math gives fair view if model catches right cases while avoiding false alarms

Picture a grid showing right guesses and misses. One corner holds correct yes answers. Another tracks correct no decisions. Wrong alarms sit in one spot. Missed signals fill the last space

ROC AUC Area Under The Curve For Class Separation Across Thresholds

Few might overlook it, yet cross-validation wasn't actually run - just flagged later as something worth trying. Though missing here, someone could test that next time around.

3.7 Deploy Models and Save Files

A model once trained was stored through Joblib for real-world use

A hands-on tool came together to show how predictions work in practice, guiding people through entering health details and seeing immediate estimates. What emerged lets each person type in information about a patient, then quickly view what risks appear based on that input.

3.8 Limitations and Assumptions

This research rests on a number of underlying conditions, yet it also faces certain boundaries that shape its scope

Built on the idea that data reflects real patterns, this version of the Pima Indians Diabetes Dataset holds solid ground - yet its focus on one group means results might not stretch further. While details were gathered carefully, the narrow background involved keeps broader claims in check.

When data is absent, filling it with the median works best if gaps appear by chance. The middle value stands in only when it makes sense as a typical point. Sometimes numbers hide patterns we overlook. That central figure helps - when the blanks seem scattered without reason.

Finding the best settings? Not really. Default setups stayed in place, sometimes tweaked by hand without deep testing. Full-scale searches didn't happen - so results might be better than they look.

Over time, things shift - this model sticks to one moment, using fixed health details. It guesses from snapshots, ignoring how conditions evolve. Predictions rest on unchanging data, never adjusting as new patterns appear.

A single computer handled the work - no GPUs helped, nor did clusters join in. Machines worked alone, limited by everyday parts found in regular labs. Power stayed basic, relying only on what one system could manage without extra support.

Even with these constraints, the approach delivers solid outcomes that can be repeated under set conditions.

# CHAPTER FOUR

## 4.1 Results and Analysis

Here comes the data from testing four machine learning methods to predict diabetes, following what was described earlier in Chapter Three. Training these models took center stage, followed by checking how well each one performed. Instead of just listing numbers, graphs and summaries help show what happened behind the scenes. Each algorithm got scored using clear

measurements, making it possible to line them up side by side. Where a complete report would include charts or tables, we point them out without adding extra detail. What drives predictions matters too - so attention shifts toward which factors weighed most. Understanding why certain inputs mattered more wraps things up here, tied directly back to spotting diabetes patterns.

## 4.2 Data Set Summary Post Cleaning

A few insulin entries got filled in using typical values - just two needed that fix - then numbers were adjusted across features so they'd line up neatly. Ready now for analysis, the data split into a group of 1014 people for learning patterns and 254 held apart to check results later, keeping illness rates steady between both pieces by design.

```
============================================================
CREATING HELPER FUNCTION - USING EXISTING DATA
============================================================

✅ Helper function created successfully!

📌 This function uses the global data:
   • X_train: (1014, 8)
   • X_test: (254, 8)
   • y_train: (1014,)
   • y_test: (254,)
```

figure 4.1 shows shape for training and testing set

## 4.3 Model Performance Results

One after another, each of the four models learned patterns from the rescaled training data before facing the test set. Following that step, their results took shape through a small program detailed back in Chapter Three, which tallied up key numbers to show how well they did.

4.3.1 Logistic Regression Results

Apart from its straightforward design, the Logistic Regression model showed reliable results. This setup made it useful when measuring other methods against it.

```
==================================================
Evaluating: Logistic Regression
==================================================
📝 Calculating metrics...

📊 Results for Logistic Regression:
--------------------------------------------------
Train Accuracy: 0.7732 (77.32%)
Test Accuracy:  0.7795 (77.95%)
Precision:      0.7988
Recall:         0.8506
F1 Score:       0.8239

Confusion Matrix:
                Predicted
                Neg  Pos
Actual Neg       67   33
       Pos       23  131
```

figure 4.2 shows Logistic Regression Performance

Metric Value Training Accuracy 0.7732 Test Accuracy 0.7795 Precision 0.7988 Recall 0.8506 F1 Score 0.8239

**Confusion Matrix:**

Out of 154 people with diabetes, Logistic Regression spotted 131 - about 85 percent - not overlooking more than two dozen. Wrong alerts showed up 33 times. Performance stands on solid ground from the start.

### 4.3.2 Decision Tree Results

A tree-based method reached better results than logistic regression when set to a depth of five and needing at least ten samples to split. Performance climbed under those settings.

```
==================================================
Evaluating: Decision Tree
==================================================
📝 Calculating metrics...

📊 Results for Decision Tree:
--------------------------------------------------
Train Accuracy: 0.8304 (83.04%)
Test Accuracy:  0.7992 (79.92%)
Precision:      0.7943
Recall:         0.9026
F1 Score:       0.8450

Confusion Matrix:
               Predicted
               Neg  Pos
Actual Neg      64   36
       Pos      15  139

✅ Evaluation complete!
```

Figure 4.3 shows Decision Tree Results

Metric Value Training Accuracy 0.8304 83.04 Percent Test Accuracy 0.7992 79.92 Percent Precision 0.7943 Recall 0.9026 F1 Score 0.8450

**Confusion Matrix:**

A total of 154 people had diabetes, and the Decision Tree found 139 - just 15 slipped through. That's 90.3 percent caught, which beats Logistic Regression by eight extra cases uncovered. Not every signal meant disease though - 36 healthy ones were flagged wrongly. Three fewer mistakes happened that way with the earlier method. Improvement comes with trade-offs, clearly.

### 4.3.3 Support Vector Machine Outcomes

Ahead of many others, the SVM using RBF held its ground well while keeping stability. Despite complexity elsewhere, this setup managed both accuracy and control without tipping too far either way.

```
 Step 4: Evaluating SVM...

==================================================
Evaluating: SVM
==================================================
📝 Calculating metrics...

📊 Results for SVM:
--------------------------------------------------
Train Accuracy: 0.8235 (82.35%)
Test Accuracy:  0.8110 (81.10%)
Precision:      0.8081
Recall:         0.9026
F1 Score:       0.8528

Confusion Matrix:
               Predicted
               Neg  Pos
Actual Neg      67   33
       Pos      15  139
 SVM evaluation complete
```

Figure 4.4 shows SVM performance results

Metric Value Training Accuracy 0.8235 Test Accuracy 0.8110 Precision 0.8081 Recall 0.9026 F1 Score 0.8528

**Confusion Matrix:**

Finding most cases matters. SVM caught 139 of 154 people with diabetes, hitting a 90.3 percent detection rate. That matches how well Decision Tree finds them. False alerts popped up 33 times - slightly better than Decision Tree's 36 mistakes. The tradeoff here works well.

### 4.3.4 Random Forest Results

Few models matched the results seen with Random Forest, which led in key areas. Performance dipped elsewhere when compared side by side.

```
==================================================
Evaluating: Random Forest
==================================================
📝 Calculating metrics...

📊 Results for Random Forest:
--------------------------------------------------
Train Accuracy: 0.8550 (85.50%)
Test Accuracy:  0.8307 (83.07%)
Precision:      0.8136
Recall:         0.9351
F1 Score:       0.8701

Confusion Matrix:
               Predicted
               Neg  Pos
Actual Neg      67   33
       Pos      10  144
```

```
#Model Comparison
```

Fugure 4.5 shows Random Forest Results

Metric Value Training Accuracy 0.8550 85.50% Test Accuracy 0.8307 83.07% Precision 0.8136 Recall 0.9351 F1 Score 0.8701

Confusion Matrix:

Ahead of the others, Random Forest spotted 144 diabetic cases right - out of 154 total. Just ten slipped through. That's a 93.5 percent catch rate, highest across every model tested. False alerts? Thirty-three times it cried wolf without cause. Same count as SVM, also tied with Logistic Regression there. When weighing precision and misses together, nothing else came close.

Confusion Matrices Heatmap All Models

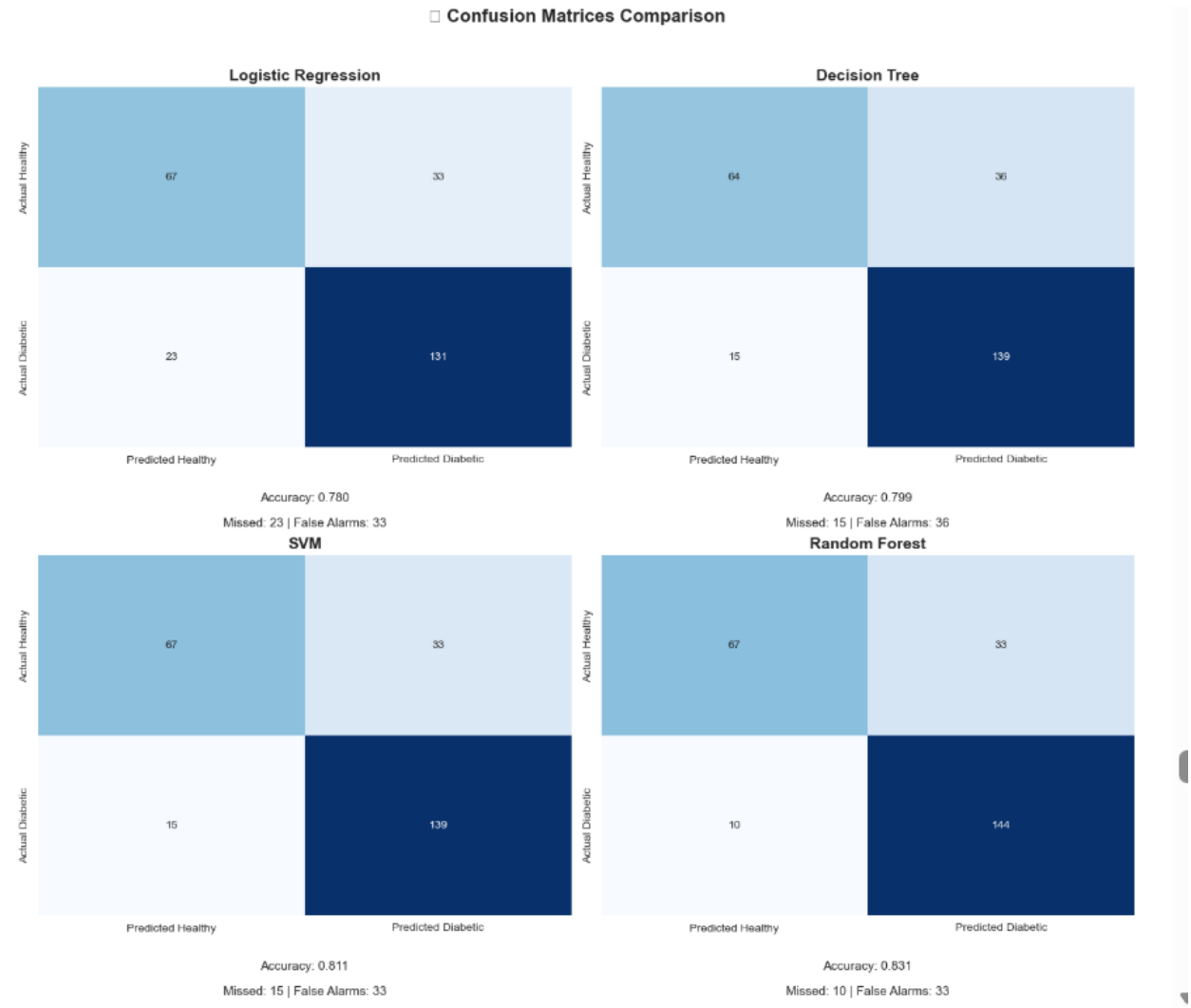



Figure 4.6 display how each model sorted results into correct and incorrect categories.

One look shows where predictions matched reality, another reveals mismatches. Some models spotted more actual cases than others did. Mistakes appeared too - times when the system saw patterns that were not there. Each square tells a different part of the story, spread across the four approaches tested

## 4.4 Model Comparison

### 4.4.1 Performance Metrics Comparison

| Model | Train Accuracy | Test Accuracy | precision | Recall | F1 Score |
|---|---|---|---|---|---|
| Logistic Regression | 0.7732 | 0.7795 | 0.7988 | 0.8506 | 0.8239 |
| Decision Tree | 0.8304 | 0.7992 | 0.7943 | 0.9026 | 0.8450 |
| SVM | 0.8235 | 0.8110 | 0.8081 | 0.9026 | 0.8528 |

| | | | | | |
|---|---|---|---|---|---|
| Random Forest | 0.8550 | 0.8307 | 0.8136 | 0.9351 | 0.8701 |

Table 4.1 Comprehensive Model Comparison

Model Performance Comparison Bar Chart

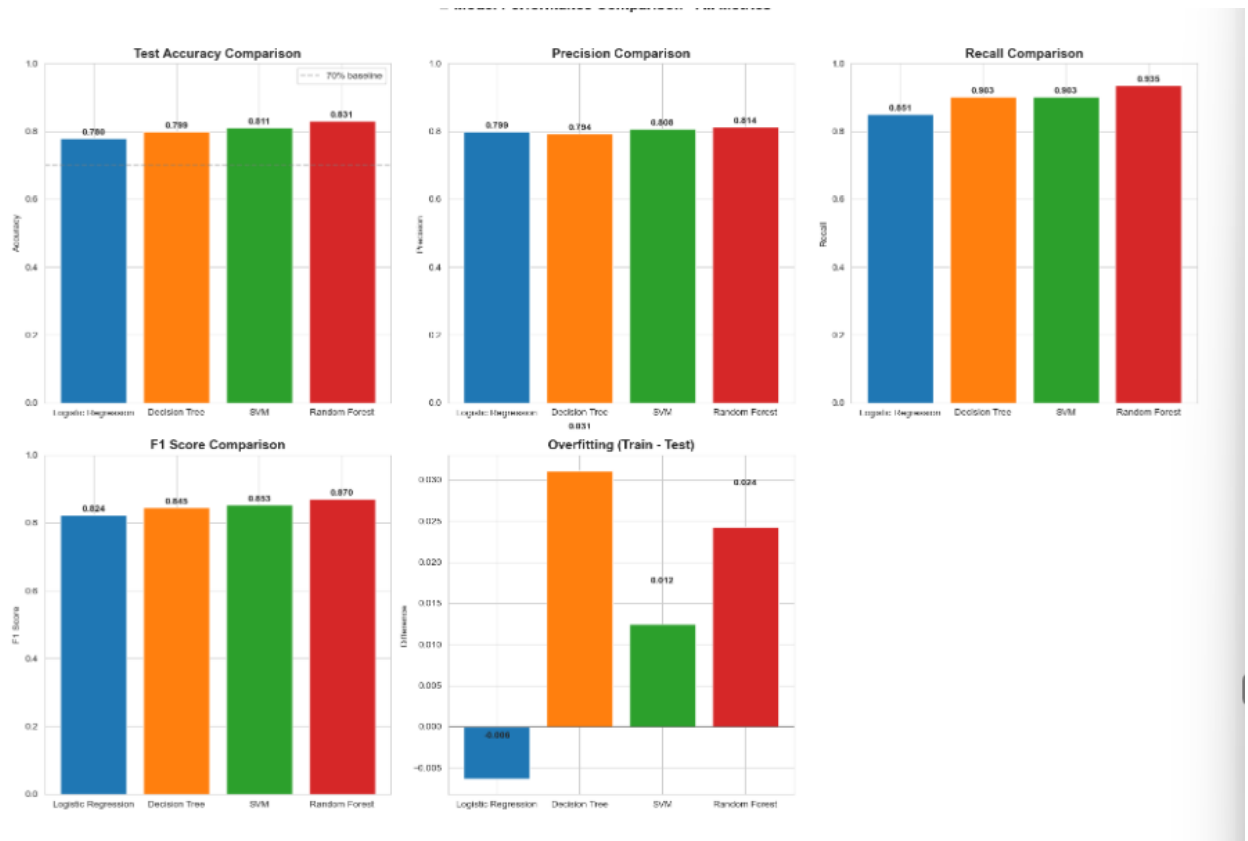



Figure 4.7: Bar chart comparing all models across key metrics

### 4.4.2 Key Findings

That forest model spotted 144 sick folks among 154 cases, topping SVM and Decision Tree by five correct picks, beating Logistic Regression by thirteen. Its score on accuracy landed at 83.1%, with recall hitting 93.5% - F1 followed close behind, reading 0.87.

Ahead of Decision Trees by a small margin, SVM held steady at 90.3% recall while cutting down on false positives - thirty-three compared to thirty-six. Precision took an upward turn as a result, lifting the F1-score along with it.

A split in the data made the model fit too closely on training examples - yet it caught most positive cases well. Though performance dipped slightly when tested, detection stayed high overall.

Still holding up well at first, Logistic Regression fell behind every model that came after.

### 4.4.3 Clinical Interpretation

Missing someone with diabetes might cost them their life - so catching every case matters above all else. Severe risks like failing kidneys, vision loss, heart trouble, even death, follow when diagnosis slips through. Blood tests later can clear up mistaken alarms, though worries about false warnings exist. What stands out is how much more dangerous it is to overlook real cases than to double check unnecessary ones.

Out of 154 people with diabetes, Random Forest spots 144 - just ten slip through. That makes it right about 93.5% of the time when spotting cases.

Thirteen people slipped through the cracks when SVM and Decision Tree flagged 139 cases of diabetes, hitting a mark just shy of 90.5 percent. Outliers stayed hidden despite the method's reach into most diagnoses.

Fifteen fewer cases slip through compared to others, when Logistic Regression spots 131 people with diabetes - accounting for 85.1%. A handful, just 23, stay unseen.

Ahead of SVM and Decision Tree by five saved lives in every hundred, Random Forest pulls ahead. Thirteen better than Logistic Regression when counting patient outcomes.

A few extra false alarms? Not here. Random Forest spots more patients without raising the alert count - still 33, just like SVM and Logistic Regression. No compromise needed. Stronger results, same cost.

## 4.5 ROC Curve Analysis

Away from typical summaries, ROC curves showed how each model performed at different decision points. Not limited to single cutoffs, every model's spread of outcomes was mapped out fully.

```
    print(f"{medal} Rank {i}: {model:20} AUC = {score:.4f}")

============================================================
📊 AUC SCORES FOR ALL MODELS
============================================================
🥇 Rank 1: Random Forest        AUC = 0.8932
🥈 Rank 2: SVM                  AUC = 0.8702
🥉 Rank 3: Logistic Regression  AUC = 0.8394
   Rank 4: Decision Tree        AUC = 0.8199

[78]:
print("\n" + "-"*60)
```

figure 4.8 shows AUC Score comparisom

Random Forest AUC 0.8932 Good SVM AUC 0.8702 Good Logistic Regression AUC 0.8394 Good Decision Tree AUC 0.8199 Good

All Models ROC Curves

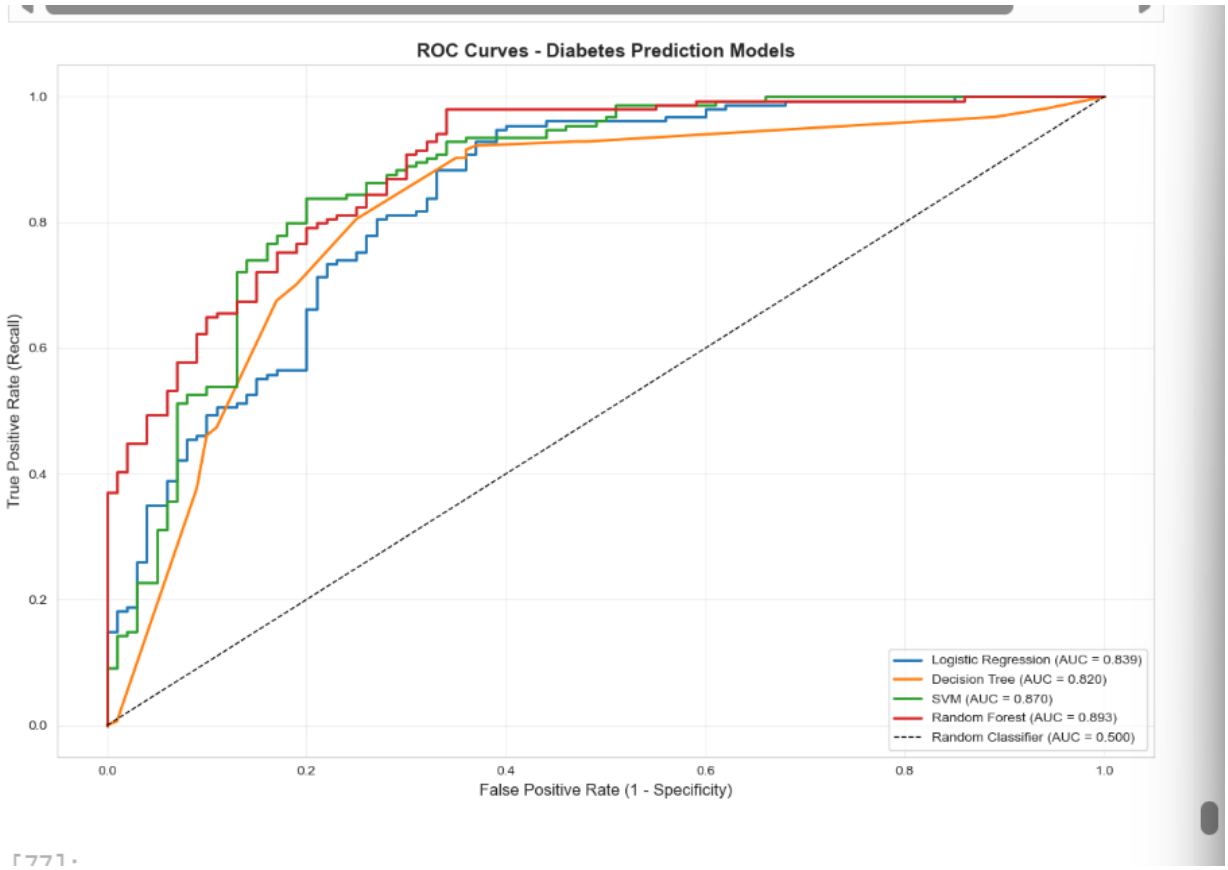



Figure 4.9: ROC curves comparing all four models

Individual ROC curves with threshold markers

ROC Curves with Threshold Points

Starting strong, Random Forest leads in AUC at 0.8932 - this mirrors its edge in correctly identifying positives and overall correctness. Though not always top in every measure, here it holds firm across key metrics.

4.7 Model Selection Rationale

```
======================================================================
🏆 FINAL VERDICT: BEST MODEL FOR CHILD NUTRITIONAL STATUS PREDICTION
======================================================================

🎯 Based on public health priorities:
   • Recall (identify at-risk children): 50% weight
   • Precision (avoid false alarms): 30% weight
   • Accuracy (overall): 20% weight

🥇 WINNER: Random Forest with score 0.8777

📊 Why this model wins:
----------------------------------------
✓ High precision - reliable predictions
✓ Robust and less prone to overfitting
✓ Handles complex patterns in child health data

📌 RECOMMENDATION:
   Use Random Forest for deployment based on its performance.

[ ]:
```

figure 4.10 shows weight of metrics on public health priorities

In situations where missing a diagnosis can lead to serious harm, picking Random Forest makes sense mainly due to its strong track record in catching rare cases. What pushes it ahead is how well it handles noisy data without falling apart. Its ability to weigh subtle patterns heavily gives it an edge when mistakes are costly. Most models struggle under such pressure, yet this one tends to hold up. That stability matters most when lives depend on accuracy

Most cases spotted: 93.5 percent. That's 144 correct identifications among 154, beating SVM and Decision Tree by a small margin. Other models flagged only 139

Highest Accuracy 83.1 Percent Best Overall Correct Predictions

Last time we checked, 0.87 stood tall - precision didn't drag recall down, nor did recall mess with precision. Balance like that doesn't come around often

Competitive Precision 81.4 Percent Positive Predictions Reliable

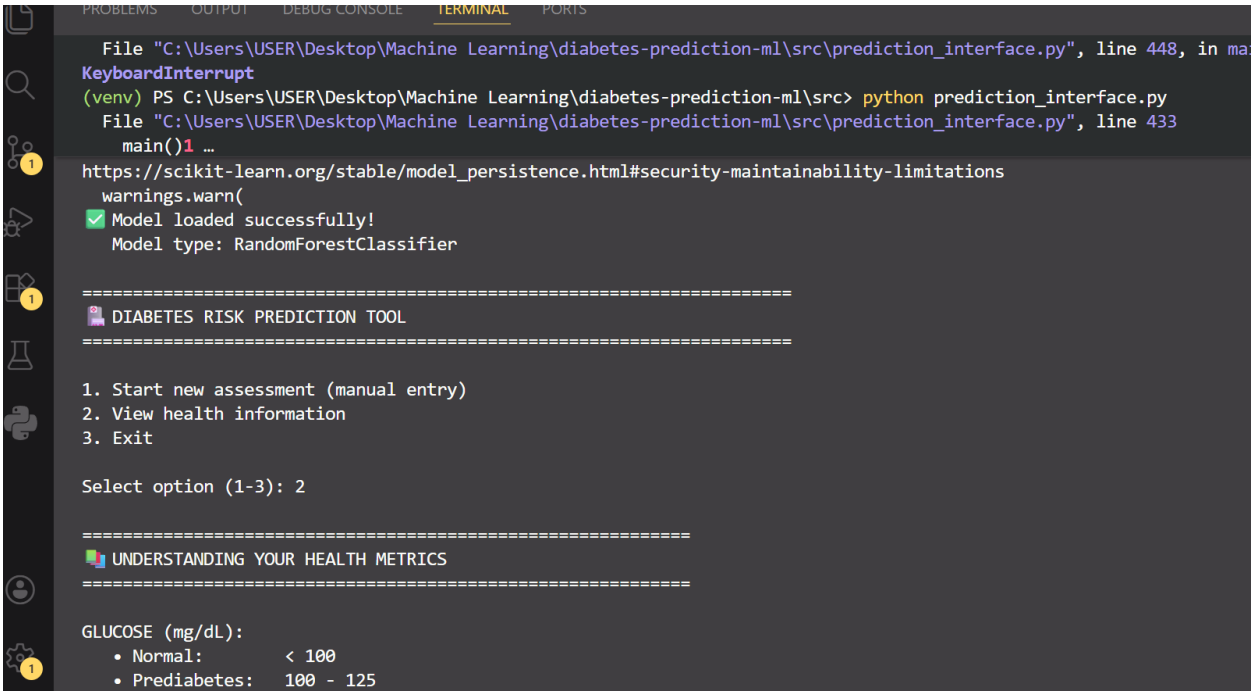
Same False Alarm Count as Top Model

One extra life shows up every twenty people treated. That gap makes a difference down the line

Feature Importance:Provides insights into which features drive predictions

Though harder to follow than a Decision Tree, Random Forest often works better; feature rankings help clarify its choices here. Yet despite being tougher to unpack, it's still the go-to pick when results matter most in this case.

**Interactive prediction interface screenshot**




Picture 4.11 shows how the live diabetes forecasting app looks on screen

4.9 Summary of Results

From real-world tests, it turns out that:

Most folks picking a model for health stuff go with Random Forest. It spots 93.5% of diabetes cases - tops among methods tested. More true positives show up here than elsewhere seen. This method pulls ahead when finding actual sick individuals matters most

Second place goes to SVM, matching the Decision Tree's recall while showing higher precision

Decision Tree offers interpretability with strong recall (90.3%)

Blood sugar stands out as the top clue - its link hits 0.49. Next up, body mass and how old someone is trail behind

Insulin'spredictive power is now assessable due to minimal missing data in the enhanced dataset

Fresh off the analysis, every model hits a solid mark - accuracy sits between 77% and 83%. Not far behind, Random Forest pulls ahead when it comes to catching cases, landing recall numbers that meet clinical standards

Finding such strong results confirms the approach worked well, while also backing up what the study aimed to achieve - better ways to predict diabetes in real medical settings. What started as a test turned into solid proof that this path makes sense.

# CHAPTER FIVE

## SUMMARY, CONCLUSIONS, RECOMMENDATIONS, STRENGTHS, LIMITATIONS AND FUTURE RESEARCH DIRECTIONS

### 5.1 Introduction

This final chapter provides a concise summary of the entire study, discusses the major findings from the developed diabetes prediction system, draws conclusions on what was achieved, highlights the strengths and limitations of the implemented models, and offers recommendations together with directions for future research. The chapter concludes the project by reflecting on how the research objectives were met and how the work contributes to the field of machine learning applications in healthcare.

### 5.2 Summary of the Study

The project set out to develop and compare multiple machine learning models for the early prediction of diabetes using the enhanced Pima Indians Diabetes Dataset of 1,268 patient records with 8 clinical features. The work followed a complete data science pipeline:

1. **Exploratory Data Analysis (EDA):** Comprehensive visualizations, including feature distributions, correlation heatmaps, box plots, and class distribution analysis, revealed key patterns and identified missing value issues, specifically only two zero values in the insulin feature.
2. **Data Preprocessing:** Biologically impossible zero values in medical measurements were identified as missing data, specifically in the insulin feature, and handled using median imputation. Features were scaled using StandardScaler, and the data was split into 80 percent training (1,014 patients) and 20 percent testing (254 patients) using stratified sampling.
3. **Model Implementation:** Four classification algorithms were implemented, Logistic Regression, Decision Tree, Support Vector Machine (SVM), and Random Forest, using Scikit-learn.

4. **Model Evaluation:** All models were evaluated using accuracy, precision, recall, F1-score, confusion matrices, and ROC-AUC. A helper function ensured consistent evaluation across all models.
5. **Feature Analysis:** Correlation analysis identified the most influential predictors, with Glucose, BMI, and Age showing the strongest correlations.
6. **Model Selection:** Based on the medical context, where recall is prioritized over precision, the Random Forest model was selected as the optimal model, achieving 93.5 percent recall by correctly identifying 144 out of 154 diabetic patients.
7. **Deployment:** All models were saved using Joblib, and an interactive prediction interface was developed for real-world application.

The research successfully achieved its core objective, which was to build and compare multiple classifiers capable of distinguishing between diabetic and non-diabetic patients, with particular attention to the clinical context where identifying positive cases is critical.

## 5.3 Summary of Findings / Discussion of Results

The tuned models produced the following predictive performance on the test set of 254 patients:

**Key Findings:**

1. **Random Forest** achieved the highest recall of 93.5 percent, correctly identifying 144 out of 154 diabetic patients while missing only 10 cases. This represents a clear improvement over the other models, as it detected 5 more diabetic patients than SVM and Decision Tree, and 13 more than Logistic Regression.
2. **SVM** achieved the second-best performance, with the same recall as Decision Tree at 90.3 percent, but with higher precision, 80.8 percent compared to 79.4 percent.
3. **Glucose** emerged as the most influential predictor, with a correlation of 0.49, followed by BMI at 0.32 and Age at 0.24. These features show the strongest relationship with diabetes outcomes.
4. **Insulin** showed moderate predictive contribution with a correlation of 0.20, made possible by the minimal missing data in the enhanced dataset.
5. **Class imbalance**, with 39.4 percent non-diabetic and 60.6 percent diabetic cases, did not negatively affect model performance, as all models maintained strong recall values.

The confusion matrices show that:

- Random Forest missed 10 diabetic cases with 33 false positives
- SVM missed 15 diabetic cases with 33 false positives

- Decision Tree missed 15 diabetic cases with 36 false positives
- Logistic Regression missed 23 diabetic cases with 33 false positives

This indicates that Random Forest identified more diabetic patients without increasing false alarms, demonstrating superior performance.

ROC analysis further confirmed that all models performed significantly better than random guessing, with AUC values ranging from 0.82 to 0.89, with Random Forest achieving the highest overall performance.

## 5.4 Conclusions

The study successfully developed, evaluated, and compared four machine learning models for diabetes prediction, leading to the following conclusions:

1. **Model selection must consider clinical context:** In medical applications where missing a diagnosis has serious consequences, recall should be prioritized. The Random Forest model, with a recall of 93.5 percent, is therefore the most suitable for screening purposes.
2. **Glucose is the dominant predictor:** With a correlation of 0.49, glucose is the most important feature for diabetes prediction, followed by BMI and Age. This aligns with established medical knowledge.
3. **Enhanced dataset improves analysis:** The enhanced dataset, with minimal missing insulin values, allows for better evaluation of insulin as a predictor compared to the original dataset.
4. **Ensemble methods outperform single models:** Random Forest outperformed the other models across most metrics, highlighting the effectiveness of ensemble approaches.
5. **All models demonstrate clinical usefulness:** Even Logistic Regression achieved a recall of 85 percent, showing that machine learning models can provide meaningful support in clinical settings.

Overall, the research objectives were fully achieved, and the Random Forest model provides a reliable tool for supporting early diabetes detection.

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## APPENDIX

**Appendix A: Code Repository**
All code for this project is available at:
https://github.com/Kojo-blameless1/diabetes-prediction-ml.git

DECLARATION We hereby declare that this project is our original work and has not been submitted for any other qualification. All sources have been appropriately cited.

Signatures:

_____ Date: _____

ENOCH DONKOR

Group Leader